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(54) **OPTICAL APPARATUS**

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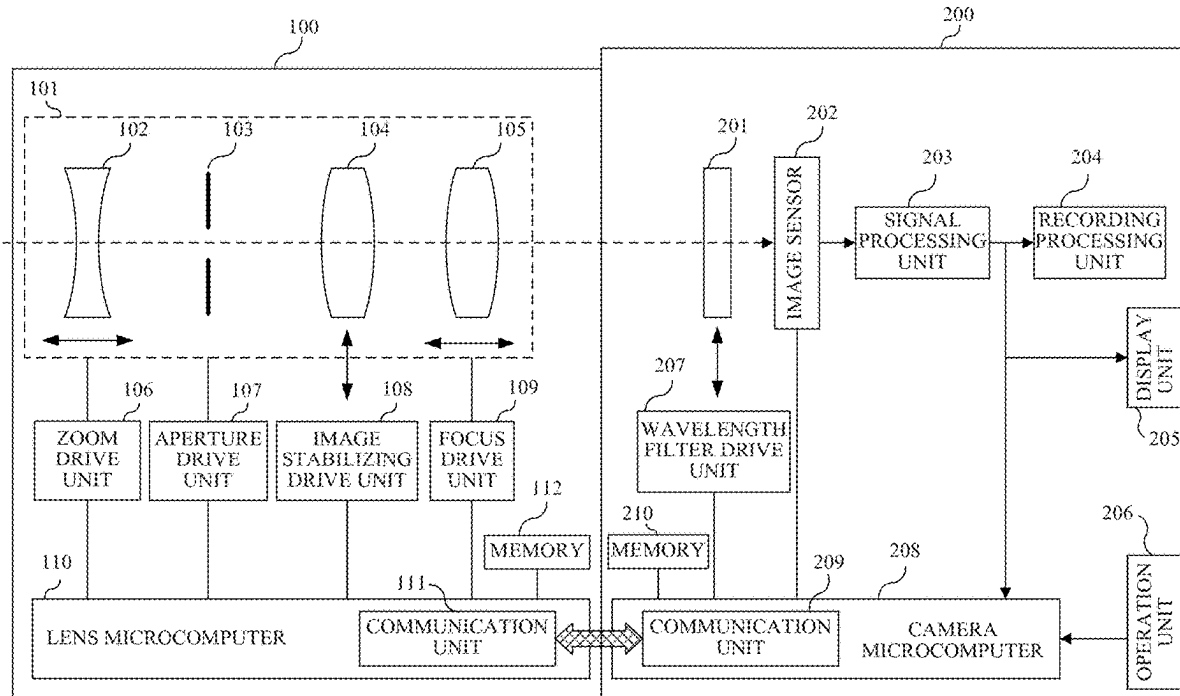
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(57)

ABSTRACT

An optical apparatus includes at least one processor that executes instructions to perform processing regarding imaging in a case where a wavelength filter configured to control a wavelength of imaging light incident on an image sensor configured to capture an object image formed by an optical system can be inserted or removed, and acquire optical information on the optical system that is used for the processing, the optical information corresponding to the wavelength of the imaging light according to at least one of an insertion or removal state of the wavelength filter, the insertion or removal state defining whether the wavelength filter is inserted into or removed from an optical axis of the optical system, and a type of the wavelength filter.



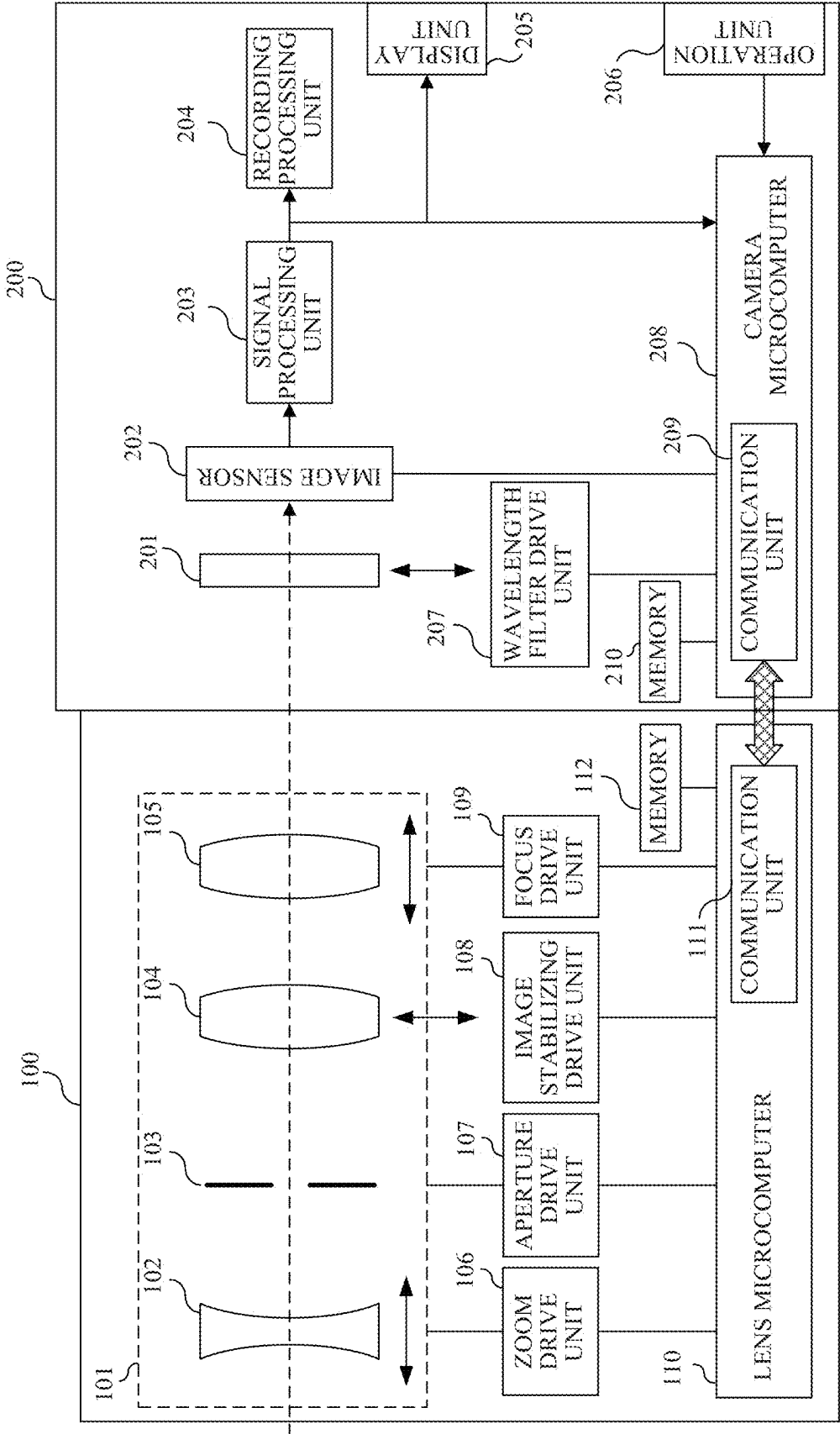


FIG. 1

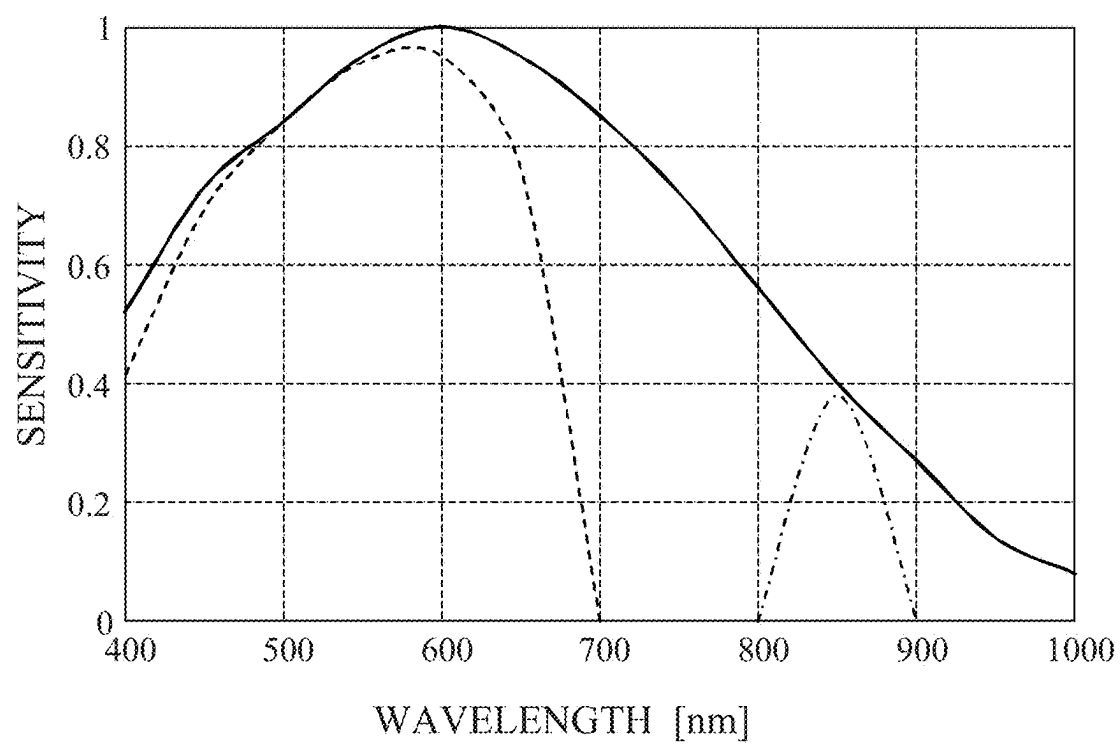


FIG. 2

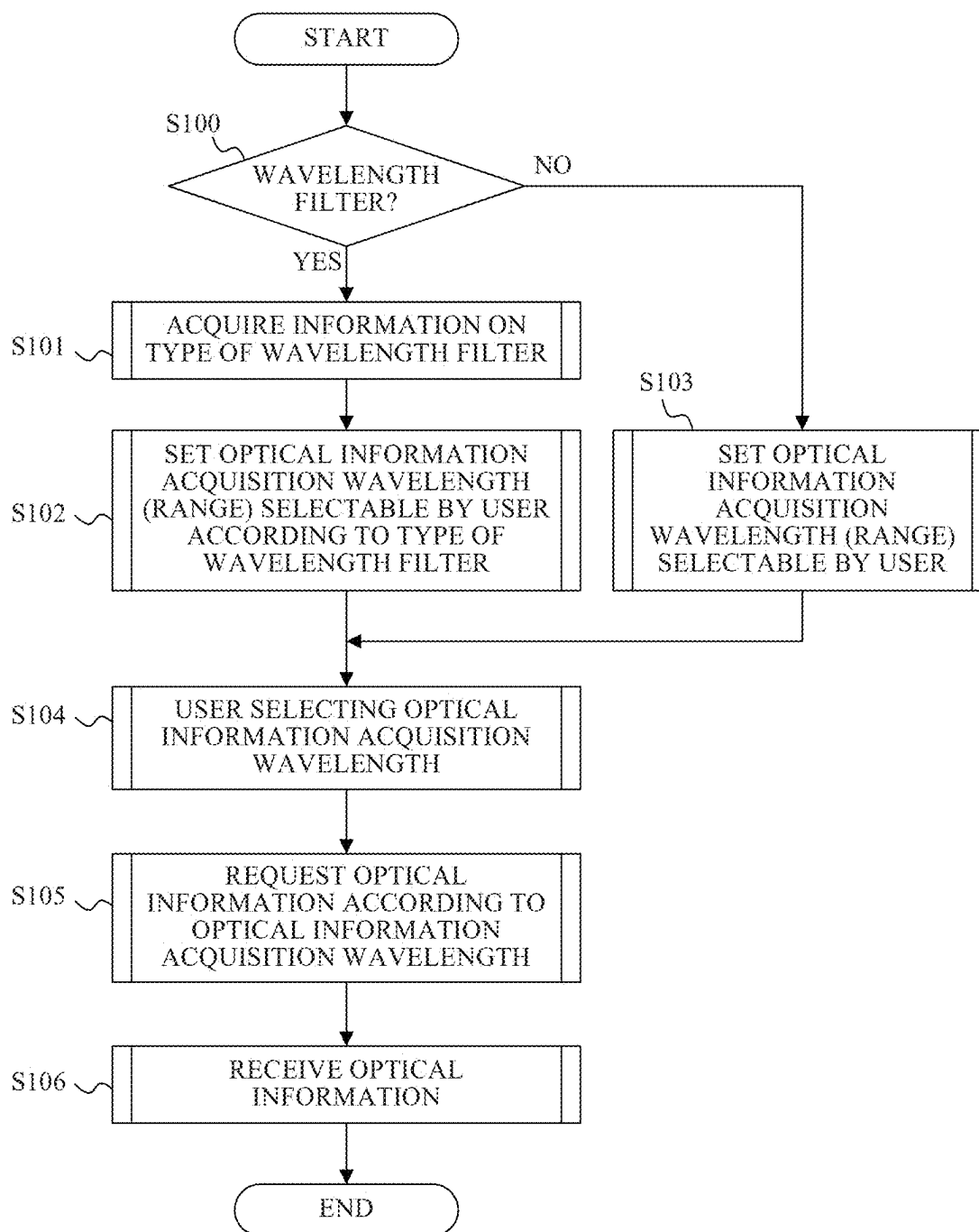


FIG. 3

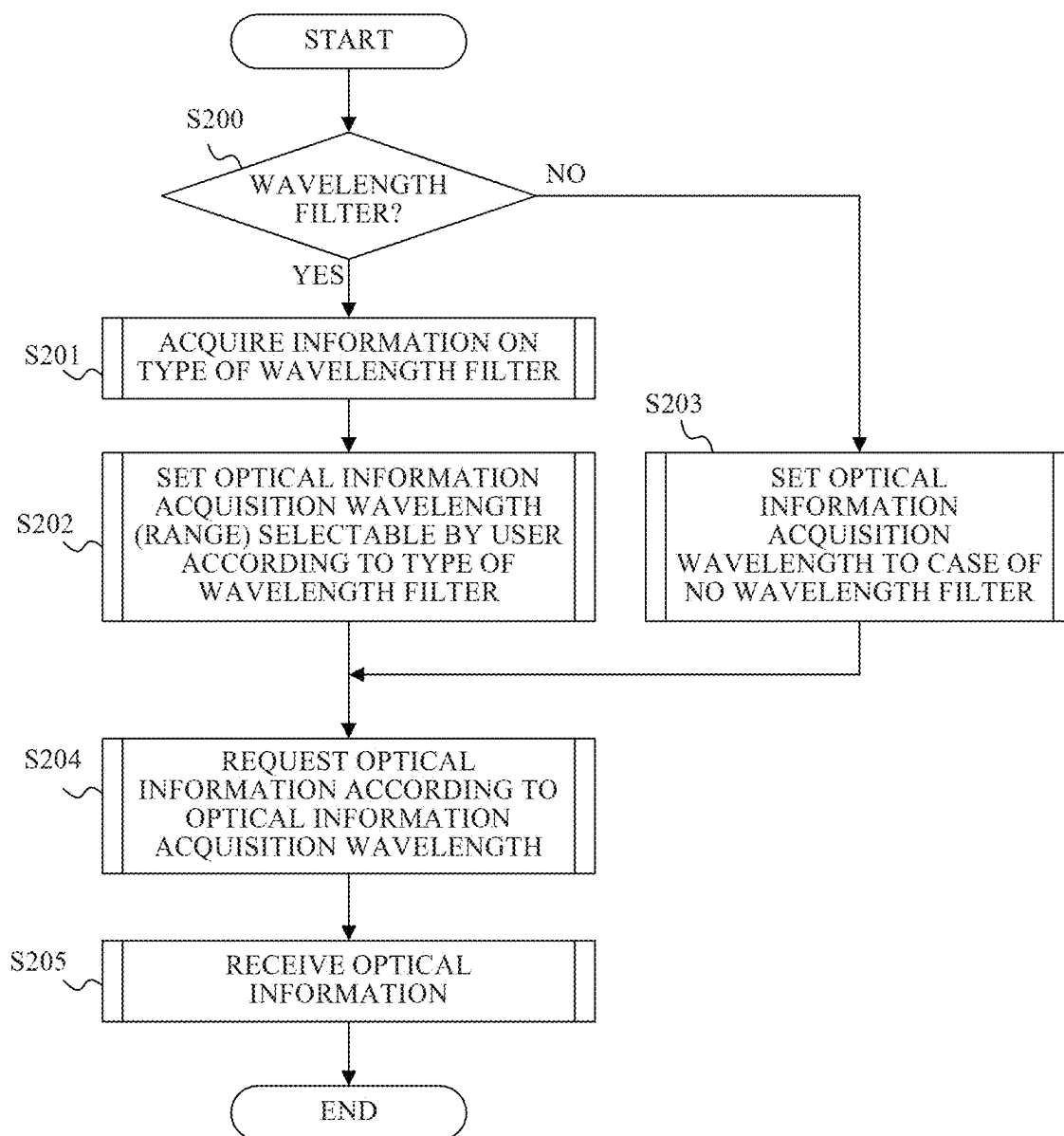


FIG. 4

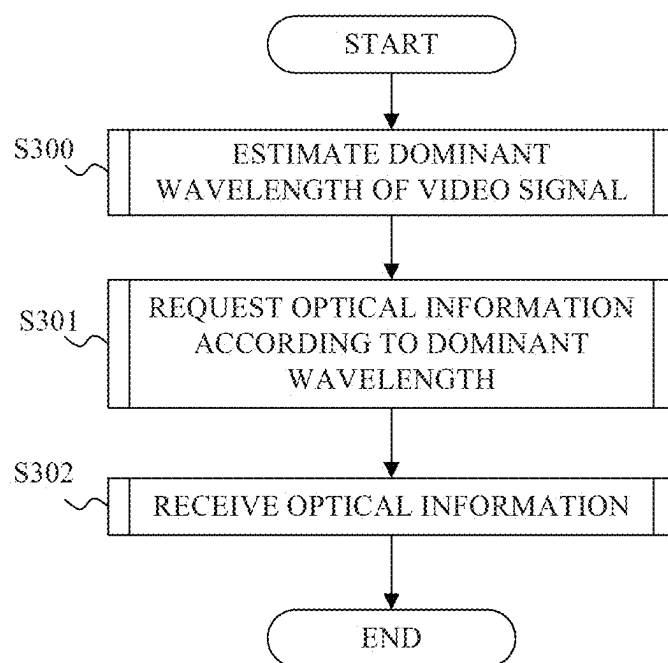


FIG. 5

OPTICAL APPARATUS

Cross-Reference to Related Applications

[0001] This application is a Continuation of International Patent Application No. PCT/JP2023/041562, filed on Nov. 20, 2023, which claims the benefit of Japanese Patent Application Nos. 2022-192589, filed on Dec. 1, 2022, and 2023-182807, filed on Oct. 24, 2023, each of which is hereby incorporated by reference herein in their entirety.

BACKGROUND

Field of the Technology

[0002] The disclosure relates to an optical apparatus, such as a lens apparatus and an image pickup apparatus.

Description of the Related Art

[0003] In an optical apparatus in which a wavelength filter configured to control the wavelength of imaging light can be inserted and removed, narrowband light such as infrared light can be imaged by inserting the wavelength filter, or broadband light including visible light and infrared light can be imaged by removing the wavelength filter.

[0004] Japanese Patent Application Laid-Open No. 2006-162757 discloses an optical apparatus in which an infrared cut filter and clear glass can be selectively inserted and removed, and a flange back adjusting lens is displaced using correction data for correcting a difference between an optical path length when the infrared cut filter is inserted and an optical path length when the clear glass is inserted.

[0005] However, in a case where optical information optimized for imaging narrowband light (such as sensitivity information for a focus lens or image stabilizing lens) is used in imaging broadband light, optical performance may deteriorate and a high-quality captured image may not be obtained.

SUMMARY

[0006] An optical apparatus according to one aspect of the disclosure includes at least one processor that executes instructions to perform processing regarding imaging in a case where a wavelength filter configured to control a wavelength of imaging light incident on an image sensor configured to capture an object image formed by an optical system can be inserted or removed, and acquire optical information on the optical system that is used for the processing, the optical information corresponding to the wavelength of the imaging light according to at least one of an insertion or removal state of the wavelength filter, the insertion or removal state defining whether the wavelength filter is inserted into or removed from an optical axis of the optical system, and a type of the wavelength filter. A control method corresponding to the above optical apparatus also constitutes another aspect of the disclosure. A storage medium storing a program that causes a computer to execute the above control method also constitutes another aspect of the disclosure.

[0007] Further features of various embodiments of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating the configuration of a lens interchangeable type camera system according to this embodiment.

[0009] FIG. 2 is a diagram illustrating spectral sensitivity characteristics according to this embodiment.

[0010] FIG. 3 is a flowchart illustrating optical information acquisition processing according to Example 1.

[0011] FIG. 4 is a flowchart illustrating optical information acquisition processing according to Example 2.

[0012] FIG. 5 is a flowchart illustrating optical information acquisition processing according to Example 3.

DESCRIPTION OF THE EMBODIMENTS

[0013] In the following, the term “unit” may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term “unit” refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or programs that, when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term “unit” refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. Depending on the specific embodiment, the term “unit” may include mechanical, optical, or electrical components, or any combination of them. The term “unit” may include active (e.g., transistors) or passive (e.g., capacitor) components. The term “unit” may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. The term “unit” may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term “unit” or “circuit” refers to any combination of the software and hardware contexts as described above. In addition, the term “element,” “assembly,” “component,” or “device” may also refer to “circuit” with or without integration with packaging materials.

[0014] Referring now to the accompanying drawings, a description will be given of embodiments according to the disclosure.

[0015] FIG. 1 illustrates the configuration of a lens interchangeable type camera system according to this embodiment. The lens interchangeable type camera system includes a camera body 200 as an image pickup apparatus (optical apparatus) and an interchangeable lens 100 as a lens apparatus that is detachably and communicably connected to the camera body 200. The interchangeable lens 100 and the camera body 200 communicate with each other via communication units 111 and 209 provided to them, respectively.

[0016] The interchangeable lens 100 includes an imaging optical system 101. The imaging optical system 101 includes a zoom lens 102, an aperture stop 103, an image stabilizing lens (image stabilizing unit) 104, and a focus lens (focusing unit) 105. The interchangeable lens 100 further includes a zoom drive unit 106, an aperture drive unit 107, an image stabilizing drive unit 108, a focus drive unit 109, a lens

microcomputer (referred to as a lens microcomputer hereinafter) **110**, and a memory **112**.

[0017] The zoom lens **102** changes a focal length of the imaging optical system **101**, i.e., varies the magnification, by moving in an optical axis direction of the imaging optical system **101** (an arrow direction in FIG. 1). The zoom drive unit **106** moves the zoom lens **102** according to an instruction from the lens microcomputer **110** that acquires the position of the zoom lens **102** detected using a position sensor such as a potentiometer (not illustrated).

[0018] The aperture stop **103** includes aperture blades that change an aperture diameter to adjust a light amount passing through the imaging optical system **101**. The aperture drive unit **107** drives an actuator such as a stepping motor to change the aperture diameter of the aperture stop **103** according to an instruction from the lens microcomputer **110** that acquires the aperture diameter detected by a sensor such as a photo-interrupter (not illustrated).

[0019] The image stabilizing lens **104** corrects image blur caused by movement applied to the camera system (camera shake) by moving in a direction orthogonal to the optical axis of the imaging optical system **101** (arrow direction in FIG. 1). The image stabilizing drive unit **108** drives an actuator such as a voice coil motor to move the image stabilizing lens **104** according to an instruction from a lens microcomputer **110** that acquires camera shake detected by a gyro sensor or acceleration sensor (not illustrated).

[0020] The focus lens **105** adjusts an imaging position of the imaging optical system **101** by moving in the optical axis direction of the imaging optical system **101**. The focus drive unit **109** drives an actuator such as a stepping motor to move the focus lens **105** according to an instruction from the lens microcomputer **110** that acquires the position of the focus lens **105** detected by a position sensor such as an encoder (not illustrated).

[0021] The memory **112** includes a Read Only Memory (ROM), a Random Access Memory (RAM), etc., and stores optical information required to control the driving of the zoom lens **102**, the aperture stop **103**, the image stabilizing lens **104**, and the focus lens **105**. Specific examples of optical information will be described later.

[0022] The camera body **200** includes a wavelength filter **201**, an image sensor **202**, a signal processing unit **203**, a recording processing unit **204**, a wavelength filter drive unit **207**, a display unit **205**, an operation unit **206**, a camera microcomputer (referred to as camera MCU hereinafter) **208**, and a memory **210**.

[0023] The image sensor **202** is a photoelectric conversion element including a CMOS sensor, a CCD sensor, or the like, and photoelectrically converts (captures) an object image formed by the imaging optical system **101**.

[0024] The wavelength filter **201** can be inserted into and removed from a filter insertion position (on the optical axis or optical path of the optical system) between the imaging optical system **101** and the image sensor **202**, and when inserted into the filter insertion position, it controls (limits) the wavelength of the imaging light that passes through the wavelength filter **201** and enters the image sensor **202**. The wavelength filter **201** is a filter such as an infrared cut filter or a bandpass filter, whose transmittance of light in a specific wavelength range is lower than that of light in other wavelength ranges. In this embodiment, the wavelength filter **201** moves inside the camera body **200** to be inserted or removed from the filter insertion position, but it may be removable to

a position outside the camera body **200**. In this embodiment, the filter insertion position is located between the imaging optical system **101** and the image sensor **202**, but the filter insertion position may be provided inside the imaging optical system.

[0025] The wavelength filter drive unit **207** inserts or removes the wavelength filter **201** into or from the filter insertion position by driving an actuator according to a manual operation by a user or an instruction from a camera microcomputer **208**. The camera microcomputer **208** can determine whether the wavelength filter **201** has been inserted or removed from the filter insertion position, and obtain information on the type of the inserted wavelength filter **201**. More specifically, the insertion or removal of the wavelength filter **201** is determined and the type information is obtained by communication with the operation history of the wavelength filter drive unit **207** and a Radio Frequency Identification (RFID) tag provided on the wavelength filter **201**. Furthermore, the insertion/removal of the wavelength filter **201** may be determined and information on the type may be obtained using a signal from a sensor that detects the insertion of the wavelength filter **201**, such as a photo-interrupter provided in the wavelength filter drive unit **207**. Moreover, information on the type of the wavelength filter **201** may be obtained based on the shape of the inserted wavelength filter **201**.

[0026] In this embodiment, the wavelength filter **201** and the wavelength filter drive unit **207** are provided in the camera body **200**, but they may be provided in the interchangeable lens **100**. In this case, the wavelength filter drive unit inserts and removes the wavelength filter **201** according to a manual operation by the user or an instruction from the lens microcomputer **110**.

[0027] An analog image signal output from the image sensor **202** that has captured an object image is input into the signal processing unit **203** and converted into a digital image signal. The signal processing unit **203** performs various signal processing such as noise removal and color correction for the digital image signal to generate a video signal, a focus signal, a luminance signal, a color difference signal, and the like. The video signal output from the signal processing unit **203** is sent to the recording processing unit **204**. The signal processing unit **203** generates still image data and moving image data from the video signal and records the data on an unillustrated recording medium.

[0028] A focus signal output from the signal processing unit **203** is input into the camera microcomputer **208**. The focus signal indicates, for example, a defocus amount in focus detection using a phase-difference detecting method. The camera microcomputer **208** converts the defocus amount into a drive amount for the focus lens **105** and transmits a focus command including the drive amount to the lens microcomputer **110**. The lens microcomputer **110** issues an instruction to the focus drive unit **109** based on the received focus command to drive the focus lens **105**. Thereby, autofocus (AF) is performed.

[0029] The luminance signal output from the signal processing unit **203** is input into the camera microcomputer **208**. The camera microcomputer **208** calculates the aperture value (F-number), shutter speed, and sensitivity of the image sensor **202** so that a brightness evaluation value obtained from the luminance signal is proper. The camera microcomputer **208** transmits an aperture command including the calculated aperture value to the lens microcomputer **110**.

The lens microcomputer **110** issues an instruction to the aperture drive unit **107** based on the received aperture command to drive the aperture stop **103**. The camera microcomputer **208** sets the calculated shutter speed and the sensitivity of the image sensor **202**. Thereby, auto-exposure (AE) is performed.

[0030] The operation unit **206** has an imaging instruction switch (not illustrated) and a switch for setting an imaging condition, etc. The camera microcomputer **208** performs various controls according to input from the operation unit **206**.

[0031] The camera microcomputer **208** performs, as an acquiring unit (processor), optical information acquisition processing to acquire optical information from the interchangeable lens **100** corresponding to a wavelength (specific wavelength: referred to as optical information acquisition wavelength hereinafter) selected by the user or automatically selected by the camera microcomputer **208**. The optical information acquisition processing will be described in detail later. The camera microcomputer **208** also performs processing regarding imaging, including control of the imaging optical system **101** using the acquired optical information and image processing to a video signal generated by the signal processing unit **203** based on the output from the image sensor **202** using the optical information. The camera microcomputer **208** and the signal processing unit **203** constitute a processing unit (processor).

[0032] FIG. 2 illustrates spectral sensitivity characteristics of light received by the image sensor **202** when no wavelength filter is inserted (solid line), when an infrared cut filter is inserted (broken line), and when a bandpass filter that transmits light with a wavelength of around 850 nm is inserted (alternate long and short dash line), into the filter insertion position.

[0033] The spectral sensitivity characteristic when the wavelength filter is removed matches the spectral sensitivity characteristic of the image sensor **202**. The spectral sensitivity characteristic when the infrared cut filter is inserted is expressed as the product of the spectral sensitivity characteristic of the infrared cut filter and the spectral sensitivity characteristic of the image sensor **202**. The spectral sensitivity characteristic when the bandpass filter that transmits light with a wavelength of around 850 nm is inserted is expressed as the product of the spectral sensitivity characteristic of the bandpass filter and the spectral sensitivity characteristic of the image sensor **202**.

[0034] In this embodiment, within the wavelength range having these spectral sensitivity characteristics, the optical information acquisition wavelength for acquiring optical information is selected by the user or automatically selected by the camera microcomputer **208**.

EXAMPLE 1

[0035] A flowchart in FIG. 3 illustrates the optical information acquisition processing that the camera microcomputer **208** executes according to a program in a case where the user selects an optical information acquisition wavelength.

[0036] In step S100, the camera microcomputer **208** determines whether or not the wavelength filter **201** is inserted into the filter insertion position. In a case where the wavelength filter **201** is inserted, the camera microcomputer **208**

proceeds to step S101, and in a case where the wavelength filter **201** is not inserted, the camera microcomputer **208** proceeds to step S103.

[0037] In step S101, the camera microcomputer **208** acquires information on the type of wavelength filter **201** inserted into the filter insertion position.

[0038] Next, in step S102, the camera microcomputer **208** sets an optical information acquisition wavelength that can be selected by the user based on the information on the type of wavelength filter **201** acquired in step S101. Table 1 illustrates an optical information acquisition wavelength for each type of wavelength filter. Types of wavelength filters include an infrared cut filter, an 850 nm bandpass filter, and a 940 nm bandpass filter. In a case where the type of wavelength filter **201** is an infrared cut filter, the selectable optical information acquisition wavelength is set to a wavelength in a range of 400 to 700 nm, and in a case where the type is an 850 nm or 940 nm bandpass filter, the optical information acquisition wavelength is set to 850 nm and 940 nm, respectively. Then, the flow proceeds to step S104.

[0039] In a case where the type of wavelength filter **201** is fixed to only an infrared cut filter, step S101 may be omitted, and the optical information acquisition wavelength of 400 to 700 nm corresponding to the infrared cut filter may be set in step S102.

[0040] On the other hand, in step S103, the camera microcomputer **208** sets an optical information acquisition wavelength selectable by the user when the wavelength filter **201** is not inserted. For example, as illustrated in Table 1, the selectable optical information acquisition wavelength is set to a wavelength in the range of 400 to 1000 nm. Then, the flow proceeds to step S104.

[0041] In step S104, the camera microcomputer **208** displays on the display unit **205** the (range of) optical information acquisition wavelength selectable by the user that has been set in step S102 or step S103.

[0042] When the user viewing the display on the display unit **205** selects an optical information acquisition wavelength via the operation unit **206**, the camera microcomputer **208** in step S105 sends an optical information transmission request including the selected optical information acquisition wavelength to the lens microcomputer **110**. The lens microcomputer **110** that has received the optical information transmission request reads out the optical information corresponding to the optical information acquisition wavelength from the memory **112** and sends it to the camera microcomputer **208**.

[0043] Tables 3 to 11 illustrate examples of optical information stored in the memory **112** corresponding to a first wavelength (400 nm) and a second wavelength (850 nm) as the optical information acquisition wavelengths. In a case where the selected optical information acquisition wavelength is 400 nm, the lens microcomputer **110** transmits optical information corresponding to 400 nm to the camera microcomputer **208**, and in a case where the selected optical information acquisition wavelength is 850 nm, the lens microcomputer **110** transmits optical information corresponding to 850 nm to the camera microcomputer **208**.

[0044] Table 3 illustrates an actual focal length for each position (zoom position) of the zoom lens **102** and each object distance, Table 4 illustrates an effective F-number for each zoom position and object distance, and Table 5 illustrates an imaging magnification for each zoom position and

object distance. Table 6 illustrates a position of the focus lens **105** at which an in-focus state is obtained for each zoom position and object distance.

[0045] Table 7 illustrates a focus sensitivity for each zoom position and object distance. The focus sensitivity is a ratio between the unit moving amount of the focus lens **105** and a change amount in the imaging position (i.e., the back focus).

[0046] Table 8 illustrates a position of a floating lens for each zoom position and object distance. Although not illustrated in FIG. 1, the floating lens is a lens that moves to reduce the change in the angle of view (breathing) and the aberration fluctuation along with the movement of the focus lens **105**.

[0047] Table 9 illustrates an image stabilizing sensitivity for each zoom position and object distance. The image stabilizing sensitivity is a ratio between the unit moving amount of the image stabilizing lens **104** and an image displacement amount on the image sensor **202** (in other words, the image stabilizing angle around the center, such as the front principal-point position).

[0048] Table 10 illustrates a front principal-point position of the imaging optical system **101** for each zoom position and object distance, and Table 11 illustrates a peripheral illumination ratio at an image height of 15 mm for each zoom position and object distance. The peripheral illumination ratio is a ratio of a light amount entering that image height to a position at an image height of 0.

TABLE 1

Type of Wavelength Filter	Wavelength Range
Infrared cut filter	400-700 [nm]
850 nm bandpass filter	850 [nm]
940 nm bandpass filter	940 [nm]
...	...
No wavelength filter	400-1000 [nm]

TABLE 2

Type of Wavelength Filter	Wavelength
Infrared cut filter	600 [nm]
850 nm bandpass filter	850 [nm]
940 nm bandpass filter	940 [nm]
...	...
No wavelength filter	700 [nm]

TABLE 3

		Object Distance		
		Infinity	...	0.3 m
Actual Focal Length [mm] (Wavelength 400 nm)				
Zoom Position	Wide	24.5	...	24.2
	...	...	...	...
	Tele	70.5	...	70
Actual Focal Length [mm] (Wavelength 850 nm)				
Zoom Position	Wide	24.6	...	24.3
	...	...	...	...
	Tele	70.8	...	70.3

TABLE 4

		Object Distance		
		Infinity	...	0.3 m
Effective Fno (Wavelength 400 nm)				
Zoom Position	Wide	4.02	...	4.04
	...	...	...	...
	Tele	4.08	...	4.15
Effective Fno (Wavelength 850 nm)				
Zoom Position	Wide	4.03	...	4.05
	...	...	...	...
	Tele	4.1	...	4.17

TABLE 5

		Object Distance		
		Infinity	...	0.3 m
Imaging Magnification (Wavelength 400 nm)				
Zoom Position	Wide	0	...	-0.6
	...	...	...	...
	Tele	0	...	-0.3
Imaging Magnification (Wavelength 850 nm)				
Zoom Position	Wide	0	...	-0.7
	...	...	...	...
	Tele	0	...	-0.2

TABLE 6

Focus Lens Position [mm] (Wavelength 400 nm)				
		Object Distance		
		Infinity	...	0.3 m
Zoom Position	Wide	1.6	...	-3.1
	...	...	...	...
	Tele	0	...	-5.7
Focus Lens Position [mm] (Wavelength 850 nm)				
		Object Distance		
		Infinity	...	0.5 m
Zoom Position	Wide	1.7	...	-3.0
	...	...	...	...
	Tele	0.3	...	-5.4

TABLE 7

		Object Distance		
		Infinity	...	0.3 m
Focus Sensitivity [mm/mm] (Wavelength 400 nm)				
Zoom Position	Wide	-0.4	...	-0.5
	...	...	...	...
	Tele	-2.1	...	-2.2
Focus Sensitivity [mm/mm] (Wavelength 850 nm)				
Zoom Position	Wide	-0.5	...	-0.6
	...	...	...	...
	Tele	-2.0	...	-2.1

TABLE 8

Floating Lens Position [mm] (Wavelength 400 nm)				
		Object Distance		
		Infinity	...	0.3 m
Zoom Position	Wide	-0.8	...	1.6
	...	...	...	...
	Tele	0	...	2.7
Floating Lens Position [mm] (Wavelength 850 nm)				
		Object Distance		
		Infinity	...	0.5 m
Zoom Position	Wide	-0.9	...	1.5
	...	...	...	...
	Tele	-0.1	...	2.5

TABLE 9

		Object Distance		
		Infinity	...	0.3 m
Image Stabilizing Sensitivity [mm/mm] (Wavelength 400 nm)				
Zoom Position	Wide	1.7	...	1.4
	...	...	...	...
	Tele	1.3	...	1.2
Image Stabilizing Sensitivity [mm/mm] (Wavelength 850 nm)				
Zoom Position	Wide	1.6	...	1.3
	...	...	...	...
	Tele	1.2	...	1.1

TABLE 10

		Object Distance		
		Infinity	...	0.3 m
Front Principal-Point Position [mm] (Wavelength 400 nm)				
Zoom Position	Wide	70	...	80
	...	...	...	...
	Tele	86	...	89
Front Principal-Point Position [mm] (Wavelength 850 nm)				
Zoom Position	Wide	71	...	81
	...	...	...	...
	Tele	87	...	91

TABLE 11

		Object Distance		
		Infinity	...	0.3 m
Peripheral Illumination Ratio at Image Height of 15 mm [%] (Wavelength 400 nm)				
Zoom Position	Wide	61	...	68
	...	...	...	...
	Tele	41	...	44

TABLE 11-continued

		Object Distance		
		Infinity	...	0.3 m
Peripheral Illumination Ratio at Image Height of 15 mm [%] (Wavelength 850 nm)				
Zoom Position	Wide	60	...	67
	...	...	...	...
	Tele	39	...	42

[0049] The optical information may be information indicating the above actual focal length, effective F-number, imaging magnification, focus lens position, focus sensitivity, floating lens position, image stabilizing sensitivity, and front principal-point position, or may be information that can be converted into each of them. In other words, the optical information may be information regarding the focal length, F-number, imaging magnification, focus lens position, focus sensitivity, floating lens position, image stabilizing sensitivity, and front principal-point position.

[0050] Next, in step S106, the camera microcomputer 208 receives the optical information transmitted from the lens microcomputer 110 and stores it in the memory 210.

[0051] Thereafter, the camera microcomputer 208 performs the above-mentioned processing to imaging using the optical information stored in the memory 210.

[0052] This embodiment performs various controls to drive the focus lens, image stabilizing lens, and aperture stop using optical information corresponding to the optical information acquisition wavelength selected by the user according to the insertion/removal or type of wavelength filter. Therefore, good optical performance can be obtained regardless of whether a wavelength filter is inserted or removed or regardless of the type of filter, and as a result, a high-quality captured image can be obtained.

[0053] This embodiment has discussed a case where the camera body 200 performs processing regarding imaging using optical information acquired from the interchangeable lens 100, but the interchangeable lens (optical apparatus) itself may perform processing regarding imaging using optical information. For example, in driving a focus lens or an image stabilizing lens based on a command from the camera body, a drive amount may be corrected based on optical information (i.e., the imaging optical system may be controlled). The processing regarding imaging also includes processing of reading optical information from the memory 112 and transmitting it to the camera body 200 configured to perform imaging. This is similarly applicable to the other embodiments described below. Also, the camera body 200 may obtain optical information via communication from a server on the cloud, rather than from the interchangeable lens 100.

EXAMPLE 2

[0054] A flowchart in FIG. 4 illustrates the optical information acquisition processing executed by the camera microcomputer 208, which automatically sets the optical information acquisition wavelength, according to a program.

[0055] In step S200, the camera microcomputer 208 determines whether or not the wavelength filter 201 is inserted into the filter insertion position. In a case where the wavelength filter 201 is inserted, the camera microcomputer 208

proceeds to step S201, and in a case where the wavelength filter 201 is not inserted, the camera microcomputer 208 proceeds to step S203.

[0056] In step S201, the camera microcomputer 208 acquires information on the type of wavelength filter 201 inserted into the filter insertion position.

[0057] Next, in step S202, the camera microcomputer 208 sets the optical information acquisition wavelength associated with the type of wavelength filter 201 based on the information on the type of wavelength filter 201 acquired in step S201. Table 2 illustrates the optical information acquisition wavelengths associated with the types of the wavelength filter. Types of wavelength filters include an infrared cut filter, an 850 nm bandpass filter, and a 940 nm bandpass filter. In a case where the type of wavelength filter 201 is an infrared cut filter, the optical information acquisition wavelength is set to 600 nm, and in a case where the type is an 850 nm or 940 nm bandpass filter, the optical information acquisition wavelength is set to 850 nm and 940 nm, respectively. Then, the flow proceeds to step S204.

[0058] On the other hand, in step S203, the camera microcomputer 208 sets the optical information acquisition wavelength to the case of no wavelength filter 201 inserted. For example, as illustrated in Table 2, the optical information acquisition wavelength is set to 700 nm. Then, the flow proceeds to step S204.

[0059] In step S204, the camera microcomputer 208 transmits an optical information transmission request including the optical information acquisition wavelength set in step S202 or step S203 to the lens microcomputer 110. The lens microcomputer 110, which has received the optical information transmission request, reads out optical information corresponding to the optical information acquisition wavelength from the memory 112 and transmits it to the camera microcomputer 208.

[0060] Next, in step S205, the camera microcomputer 208 receives the optical information sent from the lens microcomputer 110 and stores it in memory 210.

[0061] Thereafter, using the optical information stored in memory 210, the camera microcomputer 208 controls the driving of the zoom lens 102, aperture stop 103, image stabilizing lens 104, focus lens 105, floating lens, etc.

[0062] This embodiment performs various controls to drive the focus lens, image stabilizing lens, and aperture stop using optical information corresponding to the optical information acquisition wavelength that is automatically set according to the insertion/removal or type of wavelength filter. Thus, good optical performance can be obtained regardless of whether a wavelength filter is inserted or removed or regardless of the type of filter, and as a result, a high-quality captured image can be obtained.

EXAMPLE 3

[0063] A flowchart in FIG. 5 illustrates the optical information acquisition processing executed by the camera microcomputer 208 according to a program, which automatically sets the optical information acquisition wavelength according to a dominant wavelength of the light (ambient light) in an imaging environment. For example, in an imaging environment using an infrared projector, the camera microcomputer 208 sets optical information according to the dominant wavelength of infrared light as the ambient light.

[0064] In this embodiment, unlike Examples 1 and 2, the insertion/removal and type of the wavelength filter 201 are not determined, but optical information corresponding to the dominant wavelength of the ambient light received by the image sensor 202 is used according to the insertion/removal and type of the wavelength filter 201.

[0065] In step S300, the camera microcomputer 208 acquires a YUV signal expressed by a luminance signal Y and a color difference signal UV from an RGB video signal generated by imaging, and estimates the type and color temperature of the ambient light from the value obtained by accumulating the color difference signal UV. The camera microcomputer 208 further determines (estimates) the dominant wavelength of the ambient light from the estimated color temperature.

[0066] Next, in step S301, the camera microcomputer 208 sends an optical information transmission request including information on the determined dominant wavelength to the lens microcomputer 110. The lens microcomputer 110, which has received the optical information transmission request, reads out the optical information corresponding to the dominant wavelength from the memory 112 and transmits it to the camera microcomputer 208.

[0067] Next, in step S302, the camera microcomputer 208 receives the optical information sent from the lens microcomputer 110 and stores it in the memory 210.

[0068] Thereafter, using the optical information stored in the memory 210, the camera microcomputer 208 controls the driving of the zoom lens 102, aperture stop 103, image stabilizing lens 104, focus lens 105, floating lens, etc.

[0069] This embodiment performs various controls for driving the focus lens, image stabilizing lens, and aperture stop using optical information corresponding to the dominant wavelength of the ambient light according to whether the wavelength filter 201 is inserted or removed and the type of wavelength filter 201. Therefore, good optical performance can be obtained regardless of whether a wavelength filter is inserted or removed or regardless of the type of filter, and as a result, a high-quality captured image can be obtained.

OTHER EMBODIMENTS

[0070] Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the

computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disc (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0071] While the disclosure has described example embodiments, it is to be understood that the disclosure is not limited to the example embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

What is claimed is:

1. An optical apparatus comprising:
at least one processor that executes instructions to:
perform processing regarding imaging in a case where a wavelength filter configured to control a wavelength of imaging light incident on an image sensor configured to capture an object image formed by an optical system can be inserted or removed, and
acquire optical information on the optical system that is used for the processing, the optical information corresponding to the wavelength of the imaging light according to at least one of an insertion or removal state of the wavelength filter, the insertion or removal state defining whether the wavelength filter is inserted into or removed from an optical axis of the optical system, and a type of the wavelength filter.
2. The optical apparatus according to claim 1, wherein the processor is configured to:
allow a user to select a specific wavelength according to at least one of the state and the type of the wavelength filter, and
acquire the optical information corresponding to the specific wavelength.
3. The optical apparatus according to claim 2, wherein the processor is configured to display to the user a wavelength that the user can select as the specific wavelength according to at least one of the state and the type of the wavelength filter.
4. The optical apparatus according to claim 1, wherein the processor is configured to:
set a specific wavelength according to a result of determining at least one of the state and the type of the wavelength filter, and
acquire the optical information corresponding to the specific wavelength.
5. The optical apparatus according to claim 1, wherein the processor is configured to:
determine a dominant wavelength of the imaging light according to at least one of the state and the type of the wavelength filter from an output of the image sensor, and
acquire the optical information corresponding to the dominant wavelength.
6. The optical apparatus according to claim 1, further comprising a focusing unit configured to adjust an imaging position of the optical system,

wherein the optical information is used to control driving of the focusing unit.

7. The optical apparatus according to claim 1, further comprising an image stabilizing unit configured to correct image blur caused by movement applied to the optical system,

wherein the optical information is used to control driving of the image stabilizing unit.

8. The optical apparatus according to claim 1, further comprising an aperture stop configured to adjust a light amount incident on the image sensor,

wherein the optical information is used to control driving of the aperture stop.

9. The optical apparatus according to claim 1, wherein the wavelength filter is an infrared cut filter.

10. The optical apparatus according to claim 1, wherein the wavelength filter is a bandpass filter.

11. The optical information according to claim 1, wherein the optical information includes at least one of information on a focal length, information on an F-number, information on an imaging magnification, information on a position of a focus lens relative to an object distance, information on a change amount in an imaging position relative to a unit moving amount of the focus lens, information on a position of a floating lens configured to move along with movement of the focus lens, information on an image displacement amount on the image sensor relative to a unit moving amount of an image stabilizing lens, and information on a front principal-point position.

12. The optical apparatus according to claim 1, wherein the optical apparatus is an image pickup apparatus that includes the image sensor and is attachable to and detachable from a lens apparatus that includes the optical system.

13. The optical apparatus according to claim 1, wherein the optical apparatus is a lens apparatus that includes the optical system and is attachable to and detachable from an image pickup apparatus that includes the image sensor.

14. The optical apparatus according to claim 1, wherein the processor is configured to control the optical system as the processing.

15. A control method of an optical apparatus, the control method comprising:

perform processing regarding imaging in a case where a wavelength filter configured to control a wavelength of imaging light incident on an image sensor configured to capture an object image formed by an optical system can be inserted or removed; and

acquire optical information on the optical system that is used for the processing, the optical information corresponding to the wavelength of the imaging light according to at least one of an insertion or removal state of the wavelength filter, the insertion or removal state defining whether the wavelength filter is inserted into or removed from an optical axis of the optical system, and a type of the wavelength filter.

16. A storage medium storing a program that causes a computer to execute the control method according to claim 15.

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